Myntra Fashion Analytics

Decoding India's Online Fashion Market

Pricing Strategy · Discount Optimization · Category Intelligence using Data Analytics

Sector: E-Commerce / Fashion Retail

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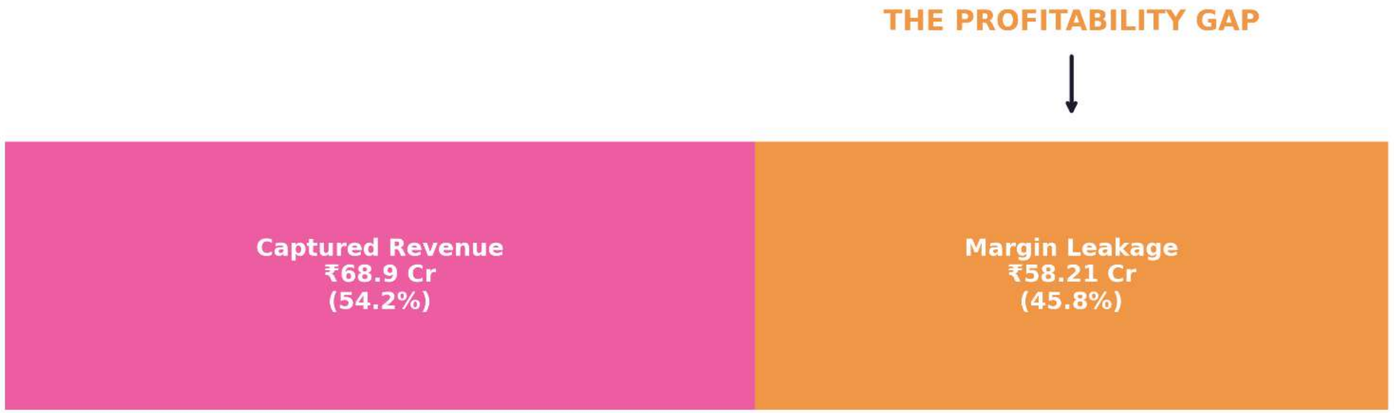
Indrustry Context

The Profitability Paradox :

High Volume, Weak Margins

Myntra generates massive top-line revenue, but aggressive discounting creates a structural disconnect between Catalogue Scale and Profit Quality.

For every Rs. 100 of Catalogue Potential, Myntra captures only Rs. 54.



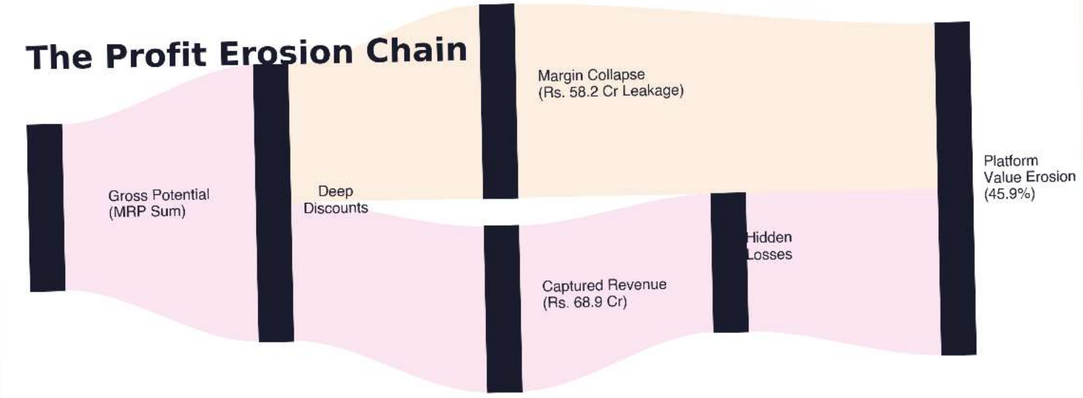
Revenue is strong (**₹68.9 Cr)**, but **45.9% of potential value** is destroyed before it hits the bottom line.

Business Context & Core Problem

Market Growth
Fashion e-commerce is scaling fast. Discounts lift volume, but unchecked promos compress margins.

Pricing Pressure
Category competition demands sharper pricing and assortment choices to protect profitability.

Stakeholders
Category leadership and pricing teams need fast, evidence-backed decisions—not intuition.



CORE BUSINESS QUESTION

How can Myntra optimize pricing, assortment, and discounts without sacrificing margins ?

Data Engineering Pipeline

PRODUCT LISTINGS	BRANDS	CATEGORIES	PLATFORM EROSION
526,564	**1,719+**	**8**	**45.9%**
catalog rows processed	normalized brand entities	core fashion segments	value erosion benchmark

Cleaning Steps

- Regex extraction for discount %
- Filled 2,300+ null discount rows
- Standardized brand and category casing
- Removed impossible values (>90% discount)
- Recomputed discount percentages
- Outlier detection and removal

DATASET COLUMNS

ProductBrand · MRP · DiscountPrice · DiscountOffer · Ratings · Reviews · Category · Gender

PROCESSING FLOW

Extract → Clean → EDA → Statistical Analysis → Dashboard

KPI Framework

Avg Discount Rate

((MRP − DP) / MRP) × 100

Primary pricing health metric

41.98%

Category Value Score

0.4×Price + 0.3×Rating + 0.3×log(Reviews)

Composite category attractiveness

Revenue Share by Gender

(Revenue by Segment / Total Revenue) × 100

Highlights demand concentration

Margin Leakage

Σ(MRP − DP) per category

Composite category attractiveness

₹58.21 Cr

Effective Revenue Index

DP × log(1 + Reviews)

Demand-weighted revenue signal

Rating–Discount Correlation

Pearson(Rating, Discount%)

Quality vs discount relationship

~0.10

High Discount SKU %

(SKUs with >50% Discount / Total SKUs) × 100

Indicates risk of brand dilution and low-quality perception

Brand Concentration Index

Top-10 brands SKU share %

Assortment dependency risk

High VFM SKU Count

Count of SKUs above VFM threshold

Identifies scalable, high-performing products

VFM Index

Price vs Rating efficiency

Identifies categories delivering strong customer value

Exploratory Analysis Findings

62.79% Revenue

Women drive nearly two-thirds of platform revenue.

₹500 – ₹1,500

The strongest price-density band across listings.

42.13 % SKUs

More than 50% discount is now structurally common.

MRP → DP

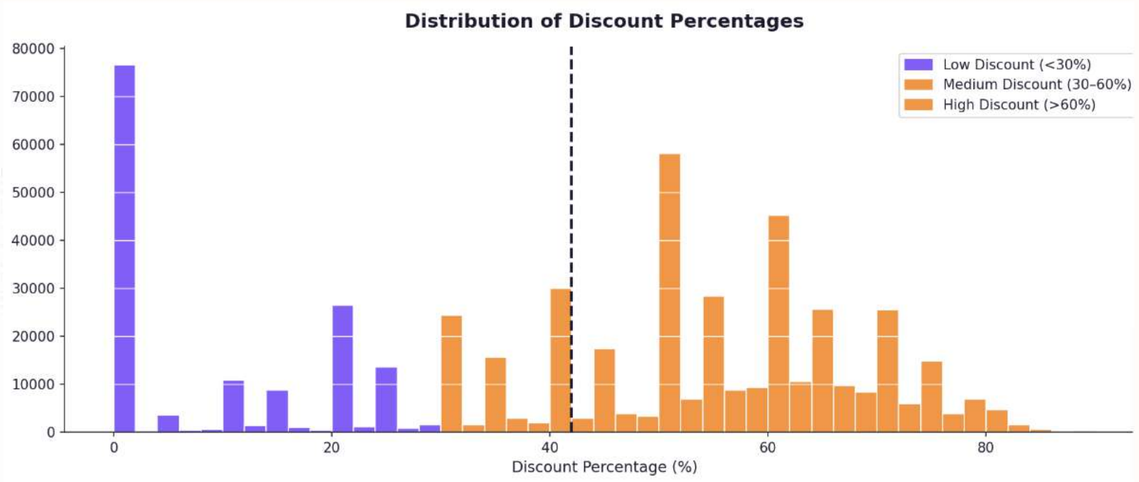
Discount price closely tracks original price levels.

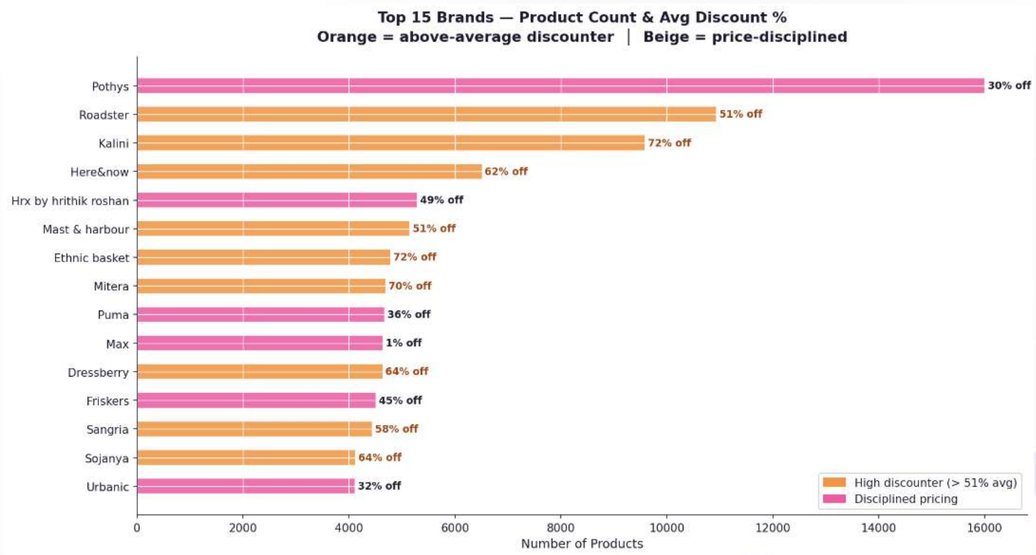
TOP 15 Brands

SKU concentration is high around a few anchor brands.

Women's Pressure

High scale comes with visible structural margin erosion.





Advanced Analysis & Root Cause Findings

METHODS USED: Pearson Correlation | OLS Regression | Category Segmentation | Brand Concentration Analysis

0.10	₹58.21 Cr	₹68.9 Cr	45.9%
DISCOUNT–RATING CORRELATION	MARGIN LEAKAGE IDENTIFIED	TOTAL REVENUE CAPTURED	PLATFORM VALUE EROSION RATE

KEY FINDINGS

- Deep discounts do **not** improve ratings — the relationship is nearly flat at r = 0.10.
- The **Golden Discount Zone** is 25–35%. Beyond 50%, profit erosion accelerates without a corresponding increase in customer ratings.
- **₹500–₹1,500 Sweet Spot**: Highest review volume and repeat purchase signals.
- **The Pareto Risk:** Top 20% of brands drive 80% of engagement; over-reliance on anchor brands.

INTERPRETATION

Price cuts aren't buying trust.

The platform is discount-heavy, but customer perception stays largely unchanged. That makes erosion structural, not promotional.

ROOT CAUSE

Massive margin leakage (**₹58.21 Cr**) is driven by the **"Price-Quality Disconnect"** — deep discounts don't fix poor product perception.

Highest-Value Insights

01

Discounts ≠ Better Ratings

r = 0.10. Heavy markdowns show no meaningful rating lift — price cuts are not improving perceived quality.

02

Women Drive Volume, Not Margin

62.79% revenue share, but the segment carries the heaviest erosion. Scale is strong; profitability is fragile.

03

₹500–₹1,500 Is the Power Band

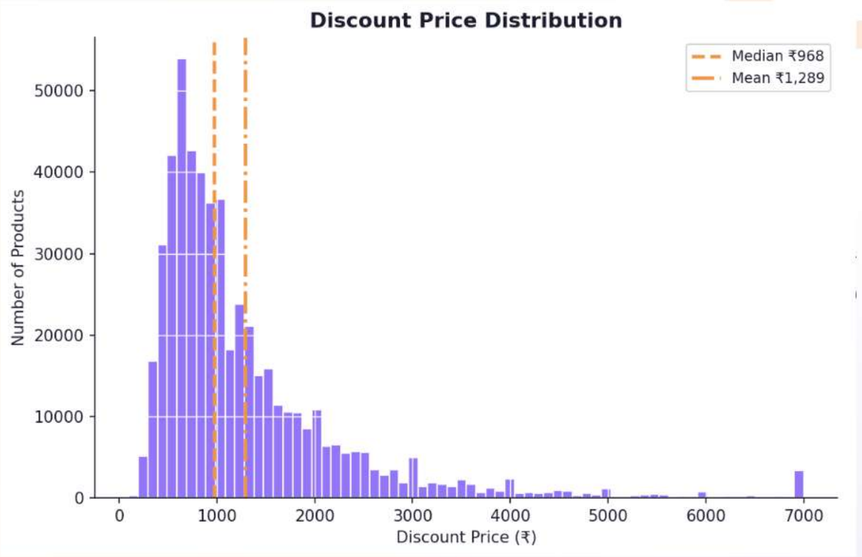
Demand clusters here. This is the sweet spot for conversion, reviews, and repeat-friendly pricing.

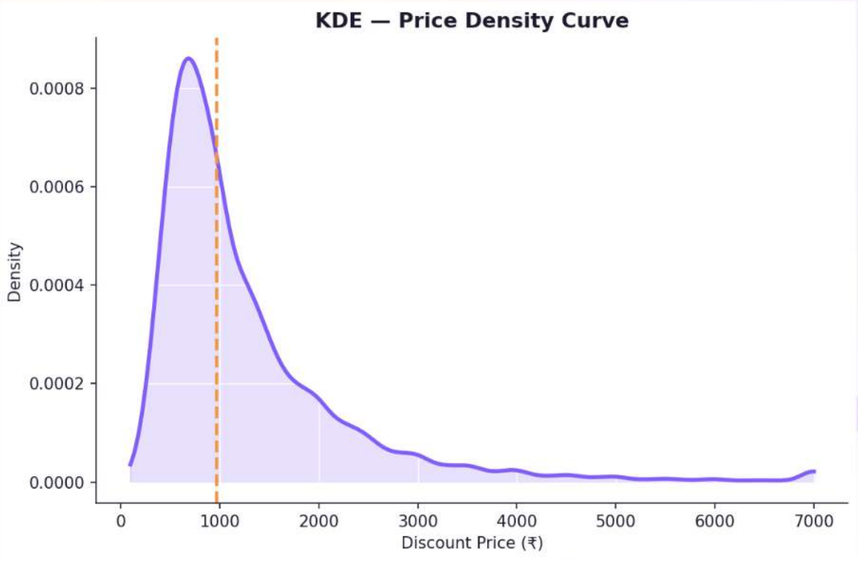
04

Top-10 Brand Dependency Risk

SKU concentration is too high. Over-reliance on anchor brands reduces assortment flexibility and negotiation power.

Kids fashion = underpenetrated, high-margin opportunity with lower competitive intensity.

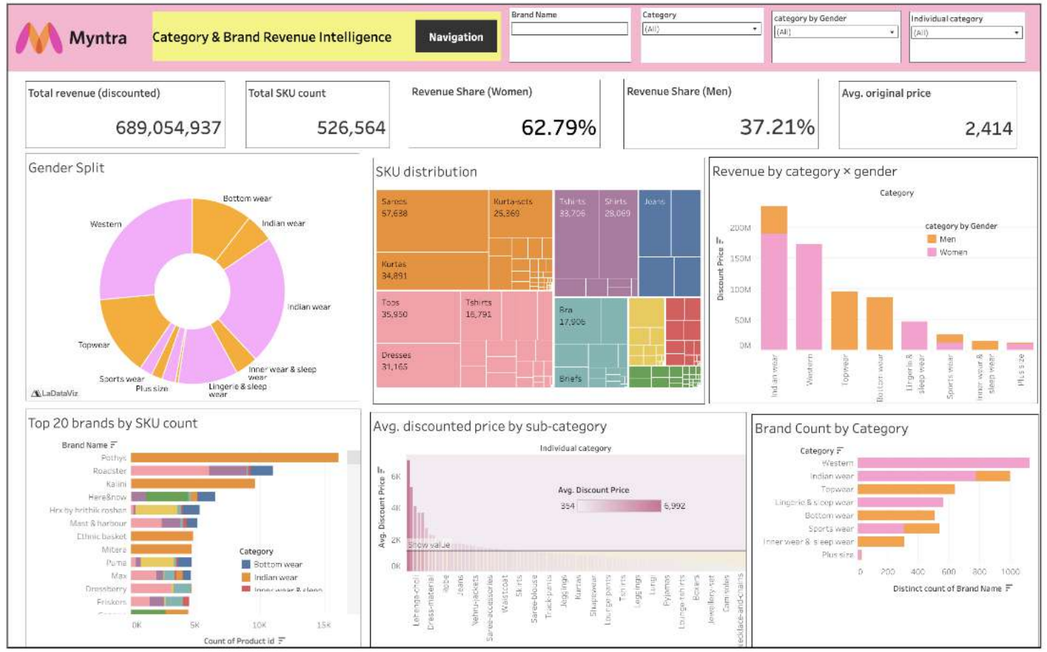




Interactive Dashboard Walkthrough

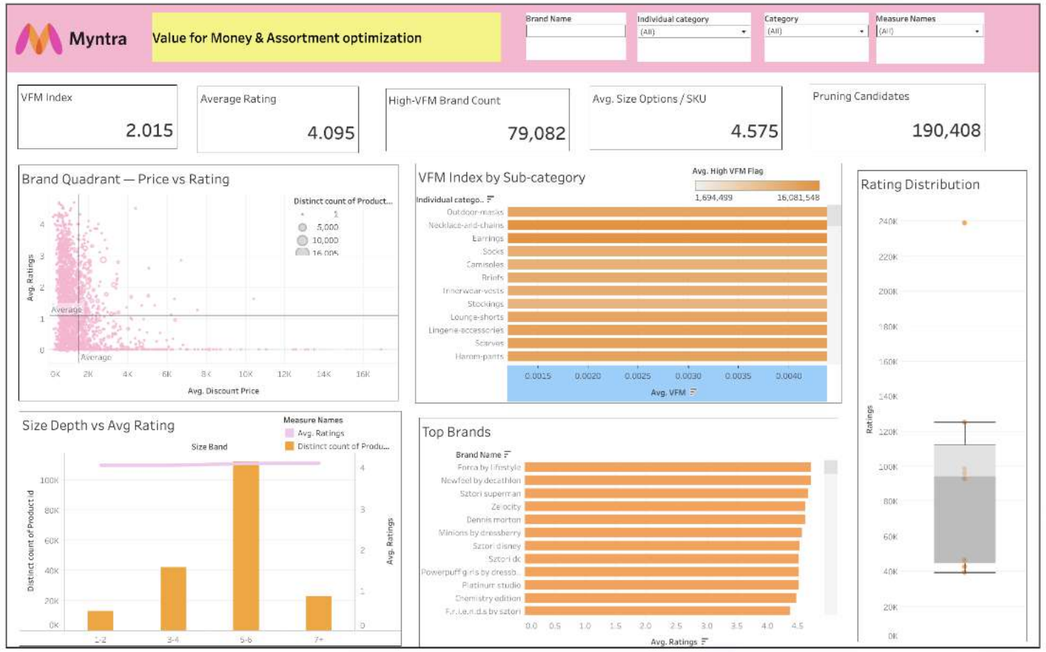
Category & Brand Revenue Intelligence

Provides a high-level view of revenue distribution across categories, brands, and gender segments, helping identify key revenue drivers and concentration areas.



Value for Money & Assortment Optimization

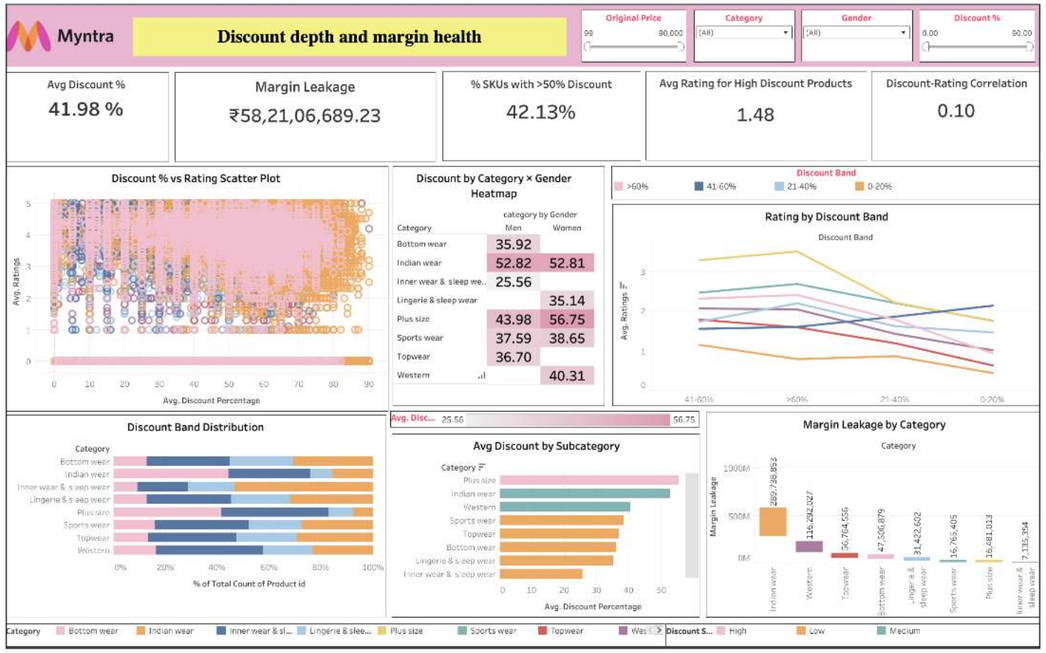
Analyzes the relationship between price, ratings, and product assortment to identify high-performing, value-driven SKUs and optimize catalog efficiency.



Interactive Dashboard Walkthrough

Discount Depth & Margin Health

Evaluates discounting patterns and their impact on margins, highlighting over-discounted segments and areas of significant margin leakage.



Strategic Recommendations

	RECOMMENDATION	ACTION	IMPACT	PRIORITY
01	**Cap Discounts at 30–40%**	Restrict >50% cuts to clearance only	**+15–25% Margin Recovery**	HIGH
02	**Shift to Value-Led Strategy**	Invest in quality and brand strength over price cuts	**Improved Ratings**	HIGH
03	**Optimize Women's Category**	Prune low-rated, high-discount SKUs	**+10–15% Profitability**	HIGH
04	**Focus on High VFM SKUs**	Increase visibility of the top 79K value SKUs	**Higher Conversions**	MED
05	**Optimize Size Depth**	Standardize around 5–6 size options per SKU	**Inventory Efficiency**	MED

Expected Business Impact

PRIORITY

HIGH

IMMEDIATE VALUE

Quick Wins

Discount caps & SKU pruning

BUILD NEXT

Strategic Bets

Kids expansion & Dynamic pricing

LOW

EASY UPGRADES

Low-Hanging Fruit

Dashboard updates & KPI monitoring

LONG HORIZON

Long-Term

ML pricing models & Competitive benchmarking

LOW — FEASIBILITY — HIGH

01

Reduced Leakage

Protect ₹58.21 Cr currently lost through value erosion.

02

Smarter Discounts

Move from blanket >50% markdowns to strategic caps.

03

Better Trust

Remove price-quality disconnects hurting customer perception.

04

Lean Assortment

Rationalize low-value SKUs in pressured categories.

05

Pricing Governance

Track VFM and leakage KPIs in real time.

Limitations

Transparent constraints behind the findings — what the analysis can signal, and what it cannot fully prove.

Data Limitations

- Dataset is catalog-level, not transaction-level (no actual sales volume)
- Reviews used as a proxy for demand, which may introduce bias
- No cost data — true net margins and profitability cannot be calculated directly.
- No time-series data → cannot analyze seasonality (sales events, festivals)

Method Limitations

- Pearson correlation captures only linear relationships (misses non-linear effects)
- VFM Index is a composite metric dependent on weighting assumptions
- Effective Revenue Index is an approximation, not actual revenue

Future Extensions

From one-time analysis to an always-on pricing intelligence system.

01

Time-Series Analytics

Track festival spikes, sale cycles, and seasonal category shifts.

02

ML Discount Optimizer

Recommend SKU-level discount bands that protect margin.

03

Dynamic Pricing Engine

Adjust prices in near real time using demand and competition signals.

04

Competitor Benchmarking

Compare Myntra against Ajio, Amazon, and Nykaa on price depth.

05

NLP Review Sentiment

Extract quality cues from review text beyond ratings alone.

EVOLUTION PATH

Descriptive Analytics

Understand what happened across pricing, discounts, categories, and margin leakage.

→

Prescriptive Pricing Intelligence

Recommend the next best pricing move for every category, brand, and SKU.

Retail success depends on profit intelligence not just revenue growth.

526K+	10+	₹58 Cr+
SKU'S ANALYZED	CRITICAL INSIGHTS	IDENTIFIED MARGIN LEAKAGE

Data analytics transforms fashion retail decision-making from **discount-driven growth** to **profit-focused optimization**.

Thank You